FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2022

<u>TEST DATE: 17-02-2022</u>

Time Allotted: 3 Hours **Maximum Marks: 300**

<u>General Instructions:</u>

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.*

***Section-B (21 – 30, 51 – 60, 81 – 90)** contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.*

Physics

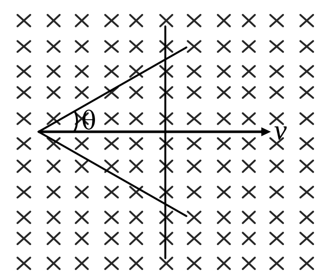
PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

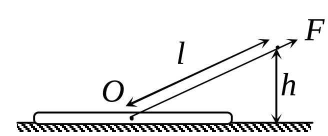
1. It is known that the temperature in the room is +20ºC when the outdoor temperature is –20ºC, and +10ºC when the outdoor temperature is – 40ºC. Then the temperature *T* of the radiator heating the room is (Assuming that radiated by the heater is proportional to the temperature difference with the room)
(A) 40ºC
(B) 60ºC
(C) 30ºC
(D) 20ºC

2. Two straight conducting rails form a right angle where their ends are joined. A conducting bar in contact with the rails start at the vertex at $t=0$ and moves with a constant velocity *v* along them as shown. A magnetic field *B* is directed into the page. The induced emf in the circuit at any time *t* is proportional to



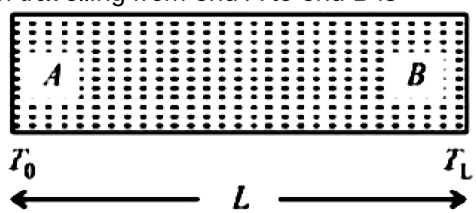
(A) v^3
(B) $\sqrt{v}$
(C) v
(D) v^2

3. A log of mass *m* is pulled at a constant velocity and with a force *F* by means of a rope of length *l*. The distance between the end of the rope and the ground is *h* as shown. The co-efficient of friction between the log and the ground is



(A) $\frac{F\sqrt{l^2-h^2}}{mgl-Fh}$
(B) $\frac{F\sqrt{l^2-h^2}}{mgl+Fh}$
(C) $\frac{Fh}{mgl-F\sqrt{l^2-h^2}}$
(D) $\frac{Fh}{mgl+F\sqrt{l^2-h^2}}$

4. The length of a mono-atomic gas in a uniform container of length L varies linearly from T_0 to T_L as shown in the figure. If the molecular weight of the gas is M_0, then the time taken by a wave pulse in travelling from end A to end B is



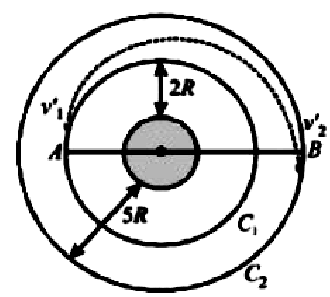
(A) $\dfrac{2L}{\left(\sqrt{T_L}+\sqrt{T_0}\right)}\sqrt{\dfrac{3M}{5R}}$

(B) $\sqrt{\dfrac{3(T_L-T_0)}{5RM_0L}}$

(C) $\dfrac{L}{(\sqrt{L_T}-\sqrt{T_0})}\sqrt{\dfrac{3M}{5R}}$

(D) $L\sqrt{\dfrac{M_0}{2R(T_L-T_0)}}$

5. An earth satellite of mass m orbits along a circular orbit C1 at a height 2R from earth's surface. It is to be transferred to a circular orbit C_2, of bigger radius, at a height 5R from earth's surface. The transfer is affected by following an elliptical path as shown in figure. Calculate the change in the energy required at point B. [R = radius of earth]



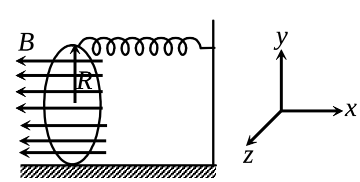
(A) $\dfrac{mgR}{36}$

(B) $\dfrac{mgR}{18}$

(C) $\dfrac{mgR}{2}$

(D) $\dfrac{mgR}{9}$

6. A ring of radius *R*, in *yz* plane, is placed in a magnetic field $B\left(-\hat{i}\right)$. Bottom point of ring is hinged and top point is attached to a spring of the spring constant *K* as shown in figure. An anticlockwise current *I* flows in the ring when viewed towards +ve *x*-axis. When ring is slightly displaced along *x*-direction it starts oscillating. Find the time Period of oscillation.



(A) $2\pi\sqrt{\dfrac{3M}{2(4K+\pi IB)}}$

(B) $\pi\sqrt{\dfrac{3M}{2(2K+\pi IB)}}$

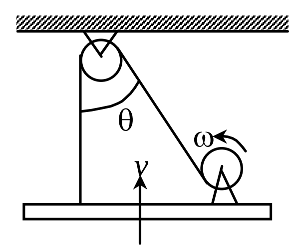
(C) $\pi\sqrt{\frac{3M}{4K+\pi IB}}$

(D) $2\pi\sqrt{\frac{3M}{4K+\pi IB}}$

7. A point source of power 4W is placed 1m below the free surface of liquid whose refractive index is '$\frac{2}{\sqrt{3}}$'. Find the rate of transfer of energy (in watt) from the liquid surface to air. Ignore any absorption or scattering of light energy.
(A) 2
(B) 1
(C) 6.5
(D) 3

8. A point light source is moving with a constant velocity *v* inside a transparent thin spherical shell of radius *R*, which is filled with a transparent liquid. If at *t* = 0 light source is at the centre of the sphere, then at what time a thin dark ring will be visible for an observer outside the sphere? The refractive index of liquid with respect to that of shell is $\sqrt{2}$.
(A) $\frac{R}{V}$
(B) $\frac{\sqrt{2}R}{V}$
(C) $\frac{R}{\sqrt{2}V}$
(D) $\frac{2R}{V}$

9. In the instant shown in the diagram, the board is moving up (vertically) with velocity *v*. The drum winds up at a constant rate ω. If the radius of the drum is *R* and the board always remains horizontal, find the value of velocity *v* in terms of *R*, θ, ω.



(A) $v=\frac{\omega R}{1+\cos\theta}$
(B) $v=\frac{\omega R}{1+\cos 2\theta}$
(C) $v=\frac{2\omega R}{1+\cos\theta}$
(D) $v=\frac{\omega R}{1+\sin\theta}$

10. A water tank having a uniform cross-sectional area *A* has an orifice of area of cross-section *a* at the bottom through which it is discharging. There is constant inflow of *Q* into the tank through another pipe. Find the time taken for the water level to rise from h_1 to h_2 if $a\sqrt{2g}=K$.

(A) $\frac{2A}{K^2}\left[Q\ \log_e\left(\frac{Q-K\sqrt{h_1}}{Q-K\sqrt{h_2}}\right)+K\ \left(\sqrt{h_1}-\sqrt{h_2}\right)\right]$

(B) $\dfrac{A}{K^2}\left[Q\ \log_e\left(\dfrac{Q-K\sqrt{h_1}}{Q-K\sqrt{h_2}}\right)+K\left(\sqrt{h_1}-\sqrt{h_2}\right)\right]$

(C) $\dfrac{A}{2K^2}\left[Q\ \log_e\left(\dfrac{Q-K\sqrt{h_1}}{Q-K\sqrt{h_2}}\right)+K\left(\sqrt{h_1}-\sqrt{h_2}\right)\right]$

(D) $\dfrac{2A}{3K^2}\left[Q\ \log_e\left(\dfrac{Q-K\sqrt{h_1}}{Q-K\sqrt{h_2}}\right)+K\left(\sqrt{h_1}-\sqrt{h_2}\right)\right]$

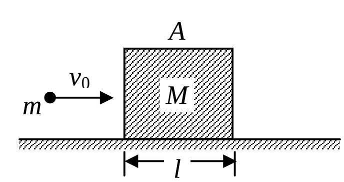
11. Suppose potential energy between electron and proton at separation *r* is given by $U = k \ln r$, where *k* is constant. For such hypothetical hydrogen atom, the ratio of energy difference between energy levels ($n = 1$ and $n = 2$) and ($n = 2$ and $n = 4$) is

(A) 1
(B) 2
(C) $\dfrac{1}{2}$
(D) 3

12. A small body starts. Sliding down an inclined plane of inclination θ, whose base length is equal to *l*. The coefficient of friction between the body and the surface is μ. If the angle θ is varied keeping *l* constant, at what angle will the time of sliding be least.

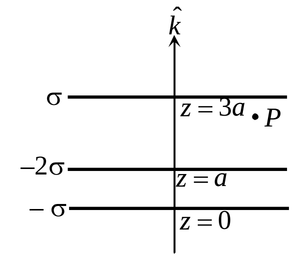
(A) $\cot 2\theta = -\dfrac{1}{\mu}$
(B) $\cot 2\theta = -\mu$
(C) $\cos 2\theta = -\dfrac{1}{\mu}$
(D) $\sin 2\theta = -\dfrac{1}{\mu}$

13. A bullet of mass *m* moving with velocity v_0 hits a wooden plank *A* of mass *M* placed on a smooth horizontal surface. The length of the plank is *l*. The bullet experiences a constant resistive force *F* inside the block. The minimum value of v_0 such that it is able to come out of the plank is



(A) $\sqrt{\dfrac{Flm}{M^2}}$
(B) $\sqrt{\dfrac{2Fl(M+m)}{Mm}}$
(C) $\sqrt{\dfrac{2Flm}{M^2}}$
(D) $\sqrt{\dfrac{Fl(M+m)}{Mm}}$

14. Three large thin sheets with surface charge density σ, –2σ and –σ are placed as shown in figure. Electric field intensity at the point *P* is



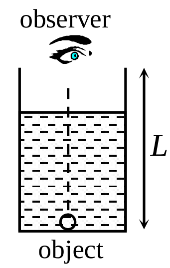
(A) $-\dfrac{4\sigma}{\varepsilon_0}\hat{k}$

(B) $\dfrac{4\sigma}{\varepsilon_0}\hat{k}$

(C) $-\dfrac{2\sigma}{\varepsilon_0}\hat{k}$

(D) $\dfrac{2\sigma}{\varepsilon_0}\hat{k}$

15. A cylinder is filled with a liquid of refractive index μ. The radius of the cylinder is decreasing at a constant rate *K*. The volume of the liquid inside the container remains constant at *V*. The observer and the object *O* are in a state of rest and at a distance *L* from each other. The apparent velocity of the object as seen by the observer, (when radius of cylinder is *r*)



(A) $\dfrac{(1-\mu)2KV}{(\pi\mu r^3)}$

(B) $\dfrac{(1-\mu)2KV}{(\pi\mu L r^2)}$

(C) $\dfrac{(1-\mu)2K}{\mu}$

(D) $\dfrac{(1-\mu)K}{2\mu}$

16. A wooden block of mass *M* rests on a horizontal surface. A bullet of mass *m* moving in the horizontal direction strikes and gets embedded in it. The combined system covers a distance *x* on the surface. If the coefficient of friction between wood and the surface is μ, the speed of the bullet at the time of striking the block is (where *m* is mass of the bullet)

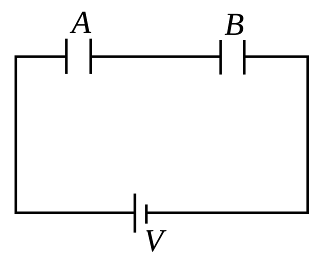
(A) $\sqrt{\dfrac{2Mg}{\mu m}}$

(B) $\sqrt{\dfrac{2\mu mg}{Mx}}$

(C) $\sqrt{2\mu gx}\left(\dfrac{M+m}{m}\right)$

(D) $\sqrt{\dfrac{2\mu mx}{M+m}}$

17. Two identical parallel plate capacitors *A* and *B* are connected in series through a battery of potential difference *V* (see figure). Area of each plate is *a* and initially plates of capacitors are separated by a distance *d*. Now, separation between plates of capacitor *B* starts increasing at constant rate *v*, find the rate by which work is done on the battery when separation between plates of capacitor *B* is 2*d*.



(A) $\dfrac{a\varepsilon_0 vV^2}{d^2}$

(B) $\dfrac{a\varepsilon_0 vV^2}{2d^2}$

(C) $\dfrac{a\varepsilon_0 vV^2}{9d^2}$

(D) $\dfrac{2a\varepsilon_0 vV^2}{d^2}$

18. Two particles *A* and *B* are separated from each other by a distance *l*. At time *t* = 0, particle *A* starts moving with uniform acceleration *a* along a line perpendicular to the initial line joining *A* and *B*. At the same moment, particle *B* starts moving with acceleration of constant magnitude *b* (> *a*) such that particle *B* always points towards the instantaneous position of *A*. The distance travelled by *B* till the moment *B* converges with *A* will be

(A) $\dfrac{b^2 l}{b^2 - a^2}$

(B) $\dfrac{a^2 l}{b^2 - a^2}$

(C) $\dfrac{(b^2 + a^2) l}{b^2 - a^2}$

(D) $\dfrac{(b^2 - a^2) l}{b^2 + a^2}$

19. If a cell produces the same amount of heat in two resistors R_1 and R_2 in the same time separately, the internal resistance of the cell is

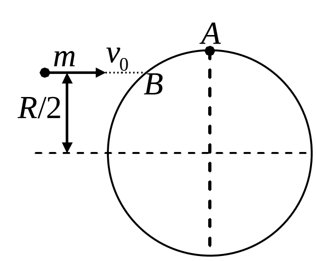
(A) $(R_1 + R_2)/2$

(B) $\sqrt{R_1\ R_2}$

(C) $\sqrt{R_1\ R_2 / 2}$

(D) $(R_1 - R_2)/2$

20. A disc of mass *m* and radius *R* is lying on a smooth horizontal surface. A particle of mass *m* moving horizontally with a velocity v_0, collides with the disc at *B* and sticks to it. Speed of the point *A* on the disc just after impact will be



(A) $\dfrac{\sqrt{31}}{8} v_0$

(B) $\dfrac{\sqrt{5}}{16} v_0$

(C) $\frac{5v_0}{16}$

(D) $\frac{v_0}{2}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

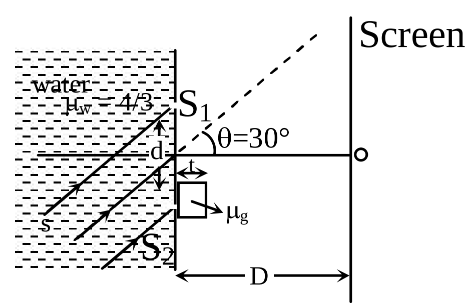
21. A hydrogen atom is in n = 5 state makes a transition to it's ground state. Assuming, H-atom is initially at rest, find the recoil speed of H-atom.

22. A police car B is chasing a culprit's car A. Car A and B are moving at constant speed V1 = 108 km/hr and V2 = 90 km/hr respectively along a straight line. The police decide to open fire and a policeman starts firing with his machine gun directly aiming at car A. The bullet has a velocity u = 305 m/s relative to the gun. The policeman keep firing for an interval of T_0 = 20 s. The Culprit experiences that the time gap between the first and the last hitting his car is Δt is

23. Sunray pass through a pinhole in the roof of a hut and produce an elliptical spot the floor. The minor and major axes of the spot are 6 cm and 12 cm respectively. The angle subtended by the diameter of the sun at our eye is 0.5°. Height of the roof in meters is

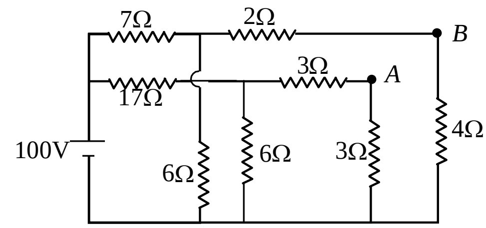
24. A railway tank wagon with its 2m diameter and 6m long horizontal cylindrical body, half full of petrol is driven around a curve of radius 100 m, at a speed of 8.33 m/s. The curve runs smoothly into a straight track and the train maintains a constant speed. If value of nf is 2.3. Find the value of 'n'. neglect viscosity and consider petrol as a solid semi cylinder sliding inside the tank. Given: $\tan^{-1}(0.07) \approx 4°$.

25. In a YDSE experiment, the two slits are covered with a transparent membrane of negligible thickness which allows light to pass through it but does not allow water. A glass slab of thickness t = 0.41 mm and refractive index μ_g = 1.5 is placed in front of one of the slits as shown in the figure. The separation between the slits is d = 0.30 mm. The entire space to the left of the slits is filled with water of refractive index μ_w = 4/3. A coherent light of intensity I and absolute wavelength λ = 5000A° is being incident on the slits making an angle 30° with horizontal. If screen is placed at a distance D = 1m from the slits, then find the distance of central maxima from O (in cm).

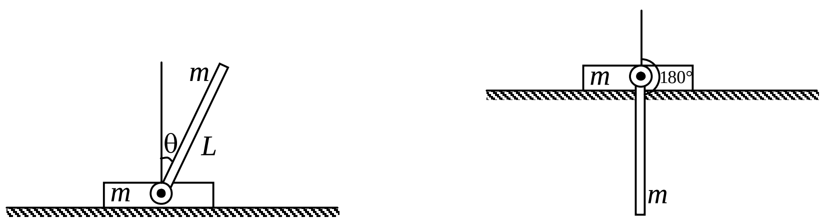


26. A metal rod AB of length 10x has its one end A in ice at 0°C, and the other end B in water at 100°C. If a point P on the rod is maintained at 400°C, then it is found that equal amounts of wate and ice evaporate and melt per unit time. The latent heat of evaporation of water is 540 cal/g and latent heat of melting of ice is 80 cal/g. If the point P is at a distance of λx from the ice end A, find the value of λ.
[Neglect any heat loss to the surrounding].

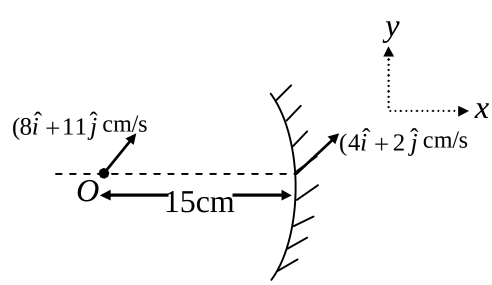
27. Determine the potential difference between points *A* and *B*, V_{AB}, in the circuit shown.



28. A rod of mass *m* and length *L* (= 50/3m) is hinged in plank of same mass *m*. The plank is kept on a smooth horizontal surface and rod makes θ with the vertical. The system is released from θ = 0°. Find the velocity (in m/s) of plank when the rod makes θ = 180°. (g = 10 m/s^2)



29. A point object located at a distance 15 cm from the pole of a concave mirror of focal length 10 cm on its principal axis is moving with a velocity $(8\hat{i} + 11\hat{j})$ cm/s and velocity of mirror is $(4\hat{i} + 2\hat{j})$ cm/s as shown. If $\vec{V}$ is the velocity of image. Then find the value of $|\vec{V}|$ (in cm/s)



30. A drop of water of mass m = 0.4 g is placed between two clean glass plates, the distance between which is 0.01 cm. Find the force of attraction between the plates. Surface tension of water = 0.08 N/m.

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

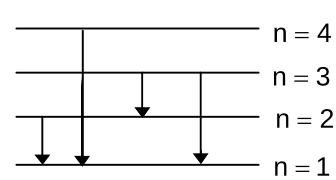
31. For a 1st order reaction
$A(g) \longrightarrow B(s) + 2C(g)$
the half life is 20 min. The reaction is carried out taking certain mass of A(g) in a closed vessel at 27°C. The initial pressure of A(g) is 200 mm. The total pressure of the reaction mixture after one hour is
(A) 550 mm
(B) 425 mm
(C) 375 mm
(D) 25 mm

32. For a first order reaction its frequency factor is $2 \times 10^{12} s^{-1}$ and threshold energy is 110.8 kJ/mole. At what temperature the half-life period will be 38.5 mins, if the average energy of reactants is 20.3 kJ/mole.
(A) 200 K
(B) 270 K
(C) 370 K
(D) 300 K

33. Which one of the following equilibria will be shifted in the backward direction upon dilution?
(A) $Cu(s) + 2Ag^{+}(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)$
(B) $HCN(aq) \rightleftharpoons H^{+}(aq) + CN^{-}(aq)$
(C) $AgCl(s) \rightleftharpoons Ag^{+}(aq) + Cl^{-}(aq)$
(D) $Zn(s) + Cu^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Cu(s)$

34. If 40 ml of 0.2 M CH_3COOH is titrated with 0.2 M NaOH. How many ml of base must be added to from a buffer solution with greatest buffer capacity?
(A) 10 ml
(B) 20 ml
(C) 30 ml
(D) 40 ml

35. Suppose that a hypothetical atom gives four transitions in its spectrum, which jump according to figure would give off second line from the red end in the spectrum?



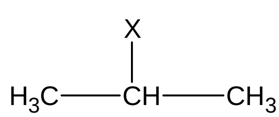
(A) 3→1
(B) 2→1
(C) 4→1
(D) 3→2

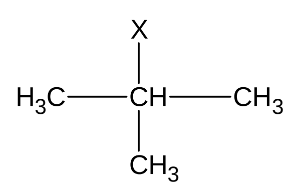
36. In which of the following compounds all the bond angles (F-F-F; where E = central element) are 90º according to VSEPR theory
(A) XeF_4
(B) XeF_6
(C) SF_6
(D) All of these

37. Transition metals are harder than s-Block metals. This is due to
(A) small size of transition metal atoms
(B) the presence of unpaired electrons in (n–1) d-orbitals in transition metals
(C) more electronegative nature of transition metals
(D) all of these

38. $Se + SeCl_4 + AlCl_3 \longrightarrow Se_8\left[AlCl_4\right]_2$
1. Comproportionation reaction
2. Al centre in $AlCl_3$ acts as lewis acid
3. Se centre in $SeCl_4$ acts as lewis base
4. stoichiometric coefficient of Se is a after balancing
Which of the above observations hold correct for the reaction given?
(A) 2 and 3
(B) 1, 2 and 3
(C) 1 and 2
(D) 2, 3 and 4

39. A compound X with molecular formula C_3H_8O can be oxidized to a compound Y with the molecular formula $C_3H_6O_2$. X is most likely to be a:
(A) primary alcohol
(B) secondary alcohol
(C) aldehyde
(D) ketone

40. Sodium tertiary butoxide forms ether only with:
(A) CH_3-CH_2-Br
(B) CH_3-X
(C)


(D)


41. The most appropriate regent for the conversion of 2-pentanone into butanoic acid is:
(A) chromic acid
(B) acidified $KMnO_4$
(C) alkaline $KMnO_4$
(D) sodium hypochlorite

42. Acetophenone can be obtained by the distillation of:
(A) $(C_6H_5COO)_2Ca$
(B) $(CH_3COO)_2Ca$
(C) $(C_6H_5COO)_2Ca$ and $(CH_3COO)_2Ca$
(D) $(C_6H_5COO)_2Ca$ and $(HCOO)_2Ca$

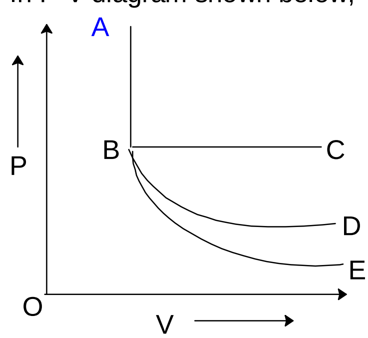
43. Which will give silver mirror test with Tollen's reagent?
(A) C_6H_5CHO
(B) CH_3-CHO
(C) HCOOH
(D) All of these

44. The number of d-electrons in Fe^{2+} (At. No. 26) is not equal to that of the
(A) p-electrons in Ne (At. No. 10)
(B) s-Electrons in Mg (At. No. 12)
(C) d-Electrons in Fe atom
(D) p-Electrons in Cl^{-} ion (At. No. 17)

45. By how many folds the temp of a gas would increase when the r.m.s. velocity of gas molecules in a closed container of fixed volume is increased from $5\times10^4 cm/s$ to 10^4 cm/s.
(A) 0.5 times
(B) 2 times
(C) 4 times
(D) 16 times

46. The solubility product of salt $AB(M.W=100)$ is $4\times10^{-10}M^2$ at 25ºC. Loss in weight of precipitate of AB by washing it with 5 litre of water is
(A) 0.1 g
(B) 1 g
(C) 10^{-4} g
(D) 0.01 g

47. In P-V diagram shown below,



(A) AB represents adiabatic process.
(B) AB represents isothermal process.
(C) AB represents isobaric process.
(D) AB represents isochoric process.

48. Which of the following is a mixed anhydride
(A) P_4O_{10}
(B) SO_3
(C) Cl_2O_6
(D) SO_2

49. In the reaction $nCH_2-CH=CH_2 \xrightarrow{[X]} \left(\begin{array}{l} -CH-CH_2- \\ \quad | \\ \quad CH_3 \end{array} \right)_n$

Reagent X is-
(A) Triethylaluminium and titanium tetrachloride
(B) Triethyl aluminium
(C) Zeigler Natta Catalyst
(D) Both (A) & (C)

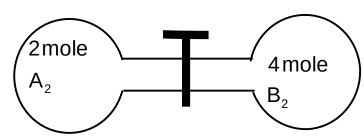
50. Which of the following is not paramagnetic-
(A) Carbon free radical
(B) Singlet carbene
(C) Triplet carbene
(D) All of these

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. There are two radioactive nuclei A & B, A is α-emitter and B is β-emitter, their decay constants are in the ratio of 1 : 2. What should be the number of atoms of each nuclei at time t = 0, so that probability of getting of α and β-particles are same at time = 0.

52. The gas A_2 in the left flask allowed to react with gas B_2, present in right flask as $A_2(g)+B_2(g) \rightleftharpoons 2AB(g);\ K_c = 4$ at $27°C$. What is the concentration of AB, when equilibrium is established? (volume of each flask is 2 litre



53. The equilibrium constant for the reaction
$CO(g)+H_2O(g) \rightleftharpoons CO_2(g)+H_2(g)$
is 4 at 1000 K
When an equimolar mixture of CO and H_2O is heated to 1000 K, the mole percentage of CO_2 formed at equilibrium is

54. $AgNO_3(s)$ is slowly added to a solution which 0.1 M in Cl^- and 0.1 M in Br^-. The % of Br^- ions precipitated when AgCl ion starts precipitating is?
$K_{sp}(AgCl)=1\times10^{-10}$ and $K_{sp}(AgBr)=1\times10^{-13}$

55. The pH of a solution obtained by mixing 2 ml of H_2SO_4 of pH = 2 and 3 ml of KOH of pH = 12 is ? (log2 = 0.3)

56. 100 ml of 2.45% (w/v) H_2SO_4 solution is mixed with 200 ml. of 7% (w/w) H_2SO_4 solution (density = 1.4 g/ml) and the mixture is dilute to 500 ml. What is the molarity of the dilute solution.

57. Element X reacts with oxygen to produce a pure sample of X_2O_3. In an experiment it is found that 1.00 g of X produces 1.16 g of X_2O_3. (Calculate the atomic weight of X. Given atomic weight of oxygen, 16.0 g mol^{-1}).

58. 0.5 mole of H_2SO_4 is mixed with 0.2 mole of $Ca(OH)_2$. The maximum number of mole of $CaSO_4$ formed is

59. The lattice parameter of Ga As (edge length of the lattice) is
(Given: Radius of Ga = $1.22\,\mathring{A}$, As $= 1.25\,\mathring{A}$)

60. AB crystallizes in a body centred cubic lattice with edge length 'a' equal to 387 pm. The distance between two oppositely changed ions in the lattice is

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If a, b, c are real numbers such that $\frac{2a+3c}{b+c}+3=0$, then the quadratic equation $ax^2 + bx + c = 0$ has
(A) atleast one root in (– 1, 0)
(B) atleast one root in (0, 1)
(C) no root in (– 1, 1)
(D) none of these

62. The domain of the function $\cos^{-1}(\log_3(x^2 + 17x + 75))$ is
(A) $(-\infty, \infty)$
(B) $[-17, 8]$
(C) $[-9, -8]$
(D) $[-9, 17]$

63. The value of $\underset{t\to 0}{\mathrm{Lt}}\frac{1}{t^2}\int_{-t}^{t}\ln(1+x)\,dx$ is equal to
(A) 1
(B) 2
(C) 0
(D) 4

64. $\int \frac{\operatorname{cosec}^2 x-2005}{\cos^{2005} x}dx$ is equal to
(A) $\frac{\cot x}{(\cos x)^{2005}}+c$
(B) $\frac{\tan x}{(\cos x)^{2005}}+c$
(C) $\frac{-(\tan x)}{(\cos x)^{2005}}+c$
(D) none of these

65. If $\alpha + i\beta = \left(\frac{-1+i\sqrt{3}}{2}\right)^{3n_1/4}(1-i)^{-2n_2}$, (where n_1 and n_2 are positive integer) then which of the following is false
(A) $\alpha = 0$ if only one of n_1 and n_2 is odd
(B) $\beta = 0$ if both n_1 and n_2 are odd
(C) $\alpha = 0$ if both n_1 and n_2 are even
(D) $\beta = 0$ if both n_1 and n_2 are even

66. z_1 and z_2 are any two distinct complex number in an argand plane. If $\alpha\beta\,|z_1| = \gamma\delta\,|z_2|$, then the complex number $\frac{\alpha\beta z_1}{\gamma\delta z_2} + \frac{\gamma\delta z_2}{\alpha\beta z_1}$ lie on the

(A) line segment [– 2, 2] on the real axis
(B) line segment [– 2, 2] of the imaginary axis
(C) unit circle $|z| = 1$
(D) none of these

67. The equation $3y^2 + 20 = 12(2x + y)$ represents a parabola with the vertex at

(A) $\left(\frac{1}{3}, 2\right)$ and axis parallel to y–axis
(B) $\left(\frac{1}{3}, 2\right)$ and focus (2, 0)
(C) $\left(2, \frac{1}{3}\right)$ and focus (2, 0)
(D) $\left(\frac{1}{3}, 2\right)$ and focus $\left(\frac{7}{3}, 2\right)$

68. The range of values of λ ($\lambda > 0$) such that the angle θ between the pair of tangents drawn from $(\lambda, 0)$ to the circle $x^2 + y^2 = 4$ lies in $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ is

(A) $\left(\frac{4}{\sqrt{3}}, 2\sqrt{2}\right)$
(B) $\left(0, \sqrt{2}\right)$
(C) (1, 2)
(D) none of these

69. If ‘a’ is a positive integer $\in$ [1, 89], then the probability that the graph of the function $f(x) = x^2 - 2(4a - 1)x + 15a^2 - 2a - 7$ is strictly above the x–axis is

(A) $\frac{1}{89}$
(B) $\frac{3}{89}$
(C) $\frac{1}{90}$
(D) none of these

70. If $\sin^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - .....\infty\right) + \cos^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - .....\infty\right) = \frac{\pi}{2}$ for $0 < |x| < \sqrt{2}$, then $\cot^{-1} x$ equal to

(A) $-\frac{\pi}{4}$
(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{3}$
(D) $\frac{\pi}{4}$

71. Solution of differential equation $y^2 + \left(\frac{dy}{dx}\right)^2 x^2 - 2xy\left(\frac{dy}{dx}\right) - \left(\frac{dy}{dx}\right)^2 = 0$ is given by
(A) $y = 2x \pm 1$
(B) $y = x \pm 2$
(C) $y = 3x \pm 2$
(D) $y = x \pm 1$

72. Let y = f(x) be a polynomial of degree 4 with real coefficients and having its zeros 1, 2, 3 only. If $f(x) > 0$ in $(-\infty, 1) \cup (3, \infty)$, then the value of $f'(1) \cdot f'(2) \cdot f'(3)$ is
(A) 0
(B) 1
(C) 3
(D) 4

73. The number of values of r satisfying the equation ${}^{69}C_{3r-1} - {}^{69}C_{r^2} = {}^{69}C_{r^2-1} - {}^{69}C_{3r}$ is
(A) 1
(B) 2
(C) 3
(D) 4

74. If $f(x) = x^4 (1 - x)^2$, $x \in [0, 1]$, then which of the following is false
(A) f(x) is continuous in [0, 1]
(B) f′(x) exist in (0, 1)
(C) $c = 2/3 \in (0, 1)$ such that f′ (c) = 0
(D) Rolle's Theorem is not applicable

75. The function f(x) = |sin 4x | + |cos 2x| is a periodic function with period
(A) $\pi/2$
(B) 2π
(C) π
(D) $\pi/4$

76. If $A = \begin{bmatrix} x & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$; $\left(x \neq \frac{-11}{3}\right)$ and det (adj (adj A)) = $(14)^4$. Then the value of x is
(A) 2
(B) –2
(C) 0
(D) none of these

77. If f (x) = $\begin{vmatrix} x+a^2 & x^4+1 & 3 \\ x+b^2 & 2x^4+2 & 3 \\ x+c^2 & 3x^4+7 & 3 \end{vmatrix}$, (where $x \neq 0$) and $f'(x) = 0$, then a^2, b^2, c^2 are in
    (A) G.P.
    (B) A.P.
    (C) H.P.
    (D) none of these

78. The lines x = ay + b, z = cy + d and x = a_1y + b_1; z = c_1y + d_1 are perpendicular if
    (A) $aa_1 + cc_1 = 1$
    (B) $aa_1 + cc_1 = -1$
    (C) $bb_1 + dd_1 = 1$
    (D) $bb_1 + dd_1 = -1$

79. The value of $\sum_{r=2}^{\infty} \tan^{-1}\left(\frac{1}{r^2-5r+7}\right)$, is
    (A) $\frac{\pi}{4}$
    (B) $\frac{\pi}{2}$
    (C) $\frac{3\pi}{4}$
    (D) $\frac{5\pi}{4}$

80. General solution of the equation $|\cos x| = \sin x$, is
    (A) $n\pi + (-1)^n \frac{\pi}{4}$
    (B) $2n\pi \pm \frac{\pi}{4}$
    (C) $n\pi + \frac{\pi}{4}$
    (D) $2n\pi + \frac{\pi}{4}$
    Where $n \in I$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If $x_k \geq 0, k = 1, 2, 3, 4, 5$ and $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ and $x_1 + x_2 + x_3 = 5$ then the number of integral solution of (x_1, x_2, x_3, x_4, x_5)

82. If $\theta = \pi/7$, then $\cos\theta + \cos 3\theta + \cos 5\theta$ is

83. The value of $\int_{10}^{100} (x-[x])dx$ is

84. If the value of the sum $29\binom{30}{0}+28\binom{30}{1}+27\binom{30}{2}+......+1\binom{30}{28}+0\binom{30}{29}-\binom{30}{30}$ where $\binom{n}{r} = {}^nC_r$, is equal to $k2^{32}$, then the value of k is equal to

85. If the line $x+y=r(r>0)$ is tangent to the curve $x^2+y^2=r$, then the value of r is

86. If the difference of the real roots of equation $x^2+4x+\frac{p}{q}=0$ is 1 (where p and q are coprime), then the value of (p + q) is

87. If $\sum_{r=0}^{n}\left(\frac{r^3+2r^2+3r+2}{(r+1)^2}\right){}^nC_r = \frac{2^2+2^3+2^4-2}{3}$, then the value of n is

88. Eccentricity of the curve represented by x = 4 (cos t + sin t) and y = $\sqrt{7}$ (cos t – sin t) is given by

89. ΔOBC has vertices O (0, 0), $B\left(\frac{a}{2}+1, \frac{a}{2}-1\right)$ and C (a, 0). If a is positive, the area of Δ OBC is 12 and D is the foot of the altitude from B to OC, then the area of triangle OBD is

90. If the vector $\vec{v}$ with magnitude $\sqrt{6}$ is along the internal bisector of the angle between $\vec{a} = 7\hat{i}-4\hat{j}-4\hat{k}$ and $\vec{b} = -2\hat{i}-\hat{j}+2\hat{k}$, then $\vec{v}\cdot\vec{a}$ equals